

Euler's Theorem

Homogeneous Function

$$u = \frac{x^4 + y^4}{x + y}$$

উপরের power

ও নিচের power

স্বতন্ত্র অব Same

অর্থাৎ Homogeneous

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = (4-1)u$$

$$= 3 \times \frac{x^4 + y^4}{x + y}$$

$$= 3u$$

$$u = f(x, y)$$

$$f(tx, ty) = t^n f(x, y) \text{ হলে}$$

সমজ্ঞক Homogeneous.

1st order

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$xu_x + yu_y = nu$$

2nd order

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n-1)u$$

Modified

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{v}{v'}$$

$v' \rightarrow$ derivative of v

$$xu_x + yu_y = n \frac{v}{v'}$$

Ex-01: $u = x^m + y^m + z^m$ show that, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2u$

→ Here, $u = x^m + y^m + z^m$

$$\frac{\partial u}{\partial x} = 2x + 0 + 0$$

$$x \frac{\partial u}{\partial x} = 2x^m \quad \text{--- (i)}$$

Again,

$$u = x^m + y^m + z^m$$

$$\frac{\partial u}{\partial y} = 0 + 2y + 0$$

$$y \frac{\partial u}{\partial y} = 2y^m \quad \text{--- (ii)}$$

Further,

$$u = x^m + y^m + z^m$$

$$\frac{\partial u}{\partial z} = 0 + 0 + 2z$$

$$z \frac{\partial u}{\partial z} = 2z^m \quad \text{--- (iii)}$$

$$\text{(i)} + \text{(ii)} + \text{(iii)} \Rightarrow$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2x^m + 2y^m + 2z^m = 2u \quad \text{(shown)}$$

Ex-03: If $u = \tan^{-1} \frac{x^3+y^3}{x+y}$ Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$

Hence, $\tan u = \frac{x^3+y^3}{x+y} = \frac{(x+y)(x^2-xy+y^2)}{(x+y)} = x^2-xy+y^2$

$$\tan u = x^2 - xy + y^2 \quad \text{--- (i)}$$

Now, Differentiate eqn (i) w.r.t x

$$\sec^2 u \frac{\partial u}{\partial x} = 2x - y + \cancel{xy} - \cancel{\frac{\partial y}{\partial x}}$$

$$\sec^2 u \left(x \frac{\partial u}{\partial x} \right) = 2x^2 - xy \quad \text{--- (ii)}$$

Again, Differentiate eqn (i) w.r.t y

$$\sec^2 u \frac{\partial u}{\partial y} = -x + 2y$$

$$\sec^2 u \left(y \frac{\partial u}{\partial y} \right) = -xy + 2y^2 \quad \text{--- (iii)}$$

$$\text{(ii)} + \text{(iii)} \Rightarrow$$

$$\sec^2 u \left(x \frac{\partial u}{\partial x} \right) + \sec^2 u \left(y \frac{\partial u}{\partial y} \right) = 2x^2 - xy - xy + 2y^2$$

$$\sec^2 u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = 2(x^2 - xy + y^2)$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \tan u \cdot \frac{1}{\sec^2 u} = 2 \times \frac{\sin u}{\cos u} \cdot \cos^2 u = 2 \sin u \cdot \cos u$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u \quad (\text{showed})$$

Ex-10: If $u = \ln \frac{x^3+y^3}{x^2+y^2}$ show by Euler's theorem

that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$.

Here,

$$e^u = \frac{x^3+y^3}{x^2+y^2} = v(x,y)$$

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 1 \times e^u$$

$$x \frac{\partial}{\partial x}(e^u) + y \frac{\partial}{\partial y}(e^u) = e^u$$

$$e^u x \frac{\partial u}{\partial x} + e^u y \frac{\partial u}{\partial y} = e^u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{e^u}{e^u} = 1 \quad (\text{Showed})$$

Ex-10(क): If $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ then show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

Here, $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$

$$u = x^0 \left[\left(\frac{y}{x} \right)^{-1} + \frac{y}{x} \times \left(\frac{z}{x} \right)^{-1} + \frac{z}{x} \right]$$

$$u = x^0 f\left(\frac{y}{x}, \frac{z}{x}\right)$$

This is 0 dimensional homogeneous function.

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0 \cdot u$$

Ex-10 (4) $u = x^3 + y^3 + z^3$ for this Homogeneous Function
Define Euler's theorem.

$$\Rightarrow u = x^3 + y^3 + z^3 \quad \text{--- (i)}$$

$$= x^3 \left(1 + \frac{y^3}{x^3} + \frac{z^3}{x^3} \right)$$

$$= x^3 f\left(\frac{y}{x}, \frac{z}{x}\right)$$

This is 3 dimensional Function.

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$$

Eqⁿ (i) Differentiate w.r.t. x ,

$$\frac{\partial u}{\partial x} = 3x^2 + 0 + 0$$

Again,

Eqⁿ (i) Differentiate w.r.t. y ,

$$x \frac{\partial u}{\partial x} = 3x^3 \quad \text{--- (ii)}$$

$$y \frac{\partial u}{\partial y} = 3y^3 \quad \text{--- (iii)}$$

Now,

$$(ii) + (iii) + (iv) \Rightarrow$$

Eqⁿ (i) Differentiate w.r.t. z ,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3(x^3 + y^3 + z^3) \quad z \frac{\partial u}{\partial z} = 3z^3 \quad \text{--- (iv)}$$

$$= 3u$$

(Proved)

Ex-10 (10) If $u = \tan^{-1} \left(\frac{x^y + y^z + z^x}{x + y + z} \right)$ then show that,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{1}{2} \sin 2u.$$

$\Rightarrow$ Hence

$$\begin{aligned} \tan u &= \frac{x^y + y^z + z^x}{x + y + z} \\ &= \frac{x^y \left(1 + \frac{y^y}{x^y} + \frac{z^y}{x^y} \right)}{x \left(1 + \frac{y}{x} + \frac{z}{x} \right)} = x^y \cdot \frac{1 + \frac{y^y}{x^y} + \frac{z^y}{x^y}}{1 + \frac{y}{x} + \frac{z}{x}} \end{aligned}$$

From Euler's theorem,

$$x \frac{\partial}{\partial x} \tan u + y \frac{\partial}{\partial y} \tan u + z \frac{\partial}{\partial z} \tan u = \tan u$$

$$x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} + z \sec^2 u \frac{\partial u}{\partial z} = \tan u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \tan u \cdot \frac{1}{\sec^2 u} = \tan u \cdot \cos^2 u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{\sin u}{\cos u} \cdot \cos^2 u$$

$$= \frac{1}{2} \cdot 2 \sin u \cdot \cos u$$

$$= \frac{1}{2} \sin 2u$$

(proved)

Ex-10 (ii) : If $f(x, y, z, w) = x^2 y^2 + z^2 w^2$ is, show that

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + w \frac{\partial f}{\partial w} = 4f$$

Hence, $f(x, y, z, w) = x^2 y^2 + z^2 w^2$

$$f(tx, ty, tz, tw) = (tx)^2 (ty)^2 + (tz)^2 (tw)^2$$

$$= t^4 x^2 y^2 + t^4 z^2 w^2$$

$$= t^4 (x^2 y^2 + z^2 w^2)$$

$$= t^4 f(x, y, z, w)$$

4 dimensional Homogenous Function.

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + w \frac{\partial f}{\partial w} = 4f \text{ (Proved)}$$

V.V.I
Ex-16

$W = f(y-x, z-y)$ is a function of x, y, z . Show that $x \frac{\partial W}{\partial x} + y \frac{\partial W}{\partial y} + z \frac{\partial W}{\partial z} = 0$

Let, $x_1 = y-x, x_2 = z-y$ then $W = f(x_1, x_2)$

$$\text{Now, } \frac{dW}{dx} = \frac{\partial W}{\partial x_1} \frac{\partial x_1}{\partial x} + \frac{\partial W}{\partial x_2} \frac{\partial x_2}{\partial x} \quad \text{--- (i)}$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial x_1} \frac{\partial x_1}{\partial y} + \frac{\partial W}{\partial x_2} \frac{\partial x_2}{\partial y} \quad \text{--- (ii)}$$

$$\frac{\partial W}{\partial z} = \frac{\partial W}{\partial x_1} \frac{\partial x_1}{\partial z} + \frac{\partial W}{\partial x_2} \frac{\partial x_2}{\partial z} \quad \text{--- (iii)}$$

Again,

$$\frac{\partial x_1}{\partial x} = \frac{\partial}{\partial x} (y - z) = -1, \quad \frac{\partial x_1}{\partial y} = 1, \quad \frac{\partial x_1}{\partial z} = 0$$

$$\frac{\partial x_2}{\partial x} = 0, \quad \frac{\partial x_2}{\partial y} = -1, \quad \frac{\partial x_2}{\partial z} = 1$$

From 1, 2 & 3

$$\frac{dw}{\partial x} = \frac{\partial w}{\partial x_1} (-1) + \frac{\partial w}{\partial x_2} (0) = -\frac{\partial w}{\partial x_1} \quad \text{--- (4)}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial x_1} \times 1 + \frac{\partial w}{\partial x_2} (-1) = \frac{\partial w}{\partial x_1} - \frac{\partial w}{\partial x_2} \quad \text{--- (5)}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial x_1} \times 0 + \frac{\partial w}{\partial x_2} \times 1 = \frac{\partial w}{\partial x_2} \quad \text{--- (6)}$$

$$\text{(4)} + \text{(5)} + \text{(6)} \Rightarrow$$

$$\frac{dw}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = -\frac{\partial w}{\partial x_1} + \frac{\partial w}{\partial x_1} - \frac{\partial w}{\partial x_2} + \frac{\partial w}{\partial x_2}$$

$$= 0$$

(proved)

Ex-17: $W = f\left(\frac{y-x}{xy}, \frac{z-y}{yz}\right)$ ज्ञात करें, कि मान क्या है,

$$x^n \frac{\partial W}{\partial x} + y^n \frac{\partial W}{\partial y} + z^n \frac{\partial W}{\partial z} = 0$$

Let $x_1 = \frac{y-x}{xy} = \frac{1}{x} - \frac{1}{y}$; $x_2 = \frac{z-y}{yz} = \frac{1}{y} - \frac{1}{z}$

Here, $W = f(x_1, x_2)$

$$\frac{\partial W}{\partial x} = \frac{\partial W}{\partial x_1} \cdot \frac{\partial x_1}{\partial x} + \frac{\partial W}{\partial x_2} \cdot \frac{\partial x_2}{\partial x} \quad \text{--- (i)}$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial x_1} \cdot \frac{\partial x_1}{\partial y} + \frac{\partial W}{\partial x_2} \cdot \frac{\partial x_2}{\partial y} \quad \text{--- (ii)}$$

$$\frac{\partial W}{\partial z} = \frac{\partial W}{\partial x_1} \cdot \frac{\partial x_1}{\partial z} + \frac{\partial W}{\partial x_2} \cdot \frac{\partial x_2}{\partial z} \quad \text{--- (iii)}$$

Again,

$$\frac{\partial x_1}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{x} - \frac{1}{y} \right) = 0; \frac{\partial x_1}{\partial y} = \frac{1}{y^2}, \frac{\partial x_2}{\partial z} = 0$$

$$\frac{\partial x_2}{\partial x} = 0, \frac{\partial x_2}{\partial y} = \frac{1}{y}, \frac{\partial x_2}{\partial z} = \frac{1}{z^2}$$

From (i) to (iii) $\Rightarrow$

$$\frac{\partial W}{\partial x} = -\frac{1}{x^2} \frac{\partial W}{\partial x_1} + 0 \Rightarrow x^n \frac{\partial W}{\partial x} = -\frac{\partial W}{\partial x_1} \quad \text{--- (iv)}$$

$$\frac{\partial W}{\partial y} = \frac{\partial W}{\partial x_1} \cdot \frac{1}{y^2} + \frac{\partial W}{\partial x_2} \cdot \frac{1}{y} \Rightarrow y^n \frac{\partial W}{\partial y} = \frac{\partial W}{\partial x_1} - \frac{\partial W}{\partial x_2} \quad \text{--- (v)}$$

$$\frac{S_w}{S_2} = 0 + \frac{1}{2^n} \frac{S_w}{S_{x_2}} \Rightarrow 2^n \frac{S_w}{S_2} = \frac{S_w}{S_{x_2}} \quad \text{--- (VI)}$$

$$\textcircled{IV} + \textcircled{V} + \textcircled{VI} \Rightarrow$$

$$2^n \frac{S_w}{S_x} + 2^n \frac{S_w}{S_y} + 2^n \frac{S_w}{S_2} = 0 \text{ (proved)}$$

Ex-07: prove that the curves $\frac{x^n}{a} + \frac{y^n}{b} = 1$ and

$\frac{x^n}{a'} + \frac{y^n}{b'} = 1$ will cut orthogonally if $a-b = a'-b'$.

$\Rightarrow$ Let, $\phi(x, y) = \frac{x^n}{a} + \frac{y^n}{b} - 1 = 0$ — (i)

$\psi(x, y) = \frac{x^n}{a'} + \frac{y^n}{b'} - 1 = 0$ — (ii)

$f_x = \frac{d}{dx} \left(\frac{x^n}{a} + \frac{y^n}{b} - 1 \right) = \frac{2x}{a}$, $f_y = \frac{2y}{b}$

and, $f_x = \frac{2x}{a'}$, $f_y = \frac{2y}{b'}$

Condition for orthogonally cut. (ଅନ୍ୟତର ଦିଗ)

$f_x \phi_x + f_y \phi_y = 0$

$\frac{2x}{a} \cdot \frac{2x}{a'} + \frac{2y}{b} \cdot \frac{2y}{b'} = 0$

$\frac{4x^2}{aa'} + \frac{4y^2}{bb'} = 0$ — (iii)

now, (i) - (ii) $\Rightarrow$

$\left(\frac{1}{a} - \frac{1}{a'} \right) x^n + \left(\frac{1}{b} - \frac{1}{b'} \right) y^n = 0$ — (iv)

from (iii) & (iv)

$\frac{1}{aa'} = \frac{1}{a} - \frac{1}{a'}$ $\frac{1}{bb'} = \frac{1}{b} - \frac{1}{b'}$

$$\frac{\left(\frac{1}{a} - \frac{1}{a'}\right)}{1/aa'} = \frac{\left(\frac{1}{b} - \frac{1}{b'}\right)}{1/bb'}$$

$$a' - a = b' - b$$

$$a - b = a' - b' \text{ (proved)}$$

Ex-11: Find the area of the triangle formed by the axes and the tangent to the curve $x^{1/3} + y^{2/3} = a^{2/3}$.

$$\Rightarrow \text{Let, } f(x, y) = x^{1/3} + y^{2/3} - a^{2/3}.$$